A LEVI-INDEX CRITERION FOR OKA COMPLEMENTS OF COMPACT DOMAINS

YUN-HENG DU

Abstract. We prove that if M is a connected Stein manifold of complex dimension at least three with the density property, then $M \setminus K$ is a connected Oka manifold for every compact $K \Subset M$ with smooth boundary such that the tangential Levi form of a defining function ρ with $K = \{\rho \leq 0\}$ has at least two nonnegative eigenvalues at every boundary point. For smooth compact domains in $\mathbb{C}^n$, our results resolve the dual Levi problem for Oka complements: pseudoconvexity is sufficient but not necessary for $n \geq 3$, and necessary but not sufficient for $n = 2$. The corresponding counterexamples occur locally on Stein manifolds with the density property in dimensions at least three and on arbitrary complex surfaces in dimension two.

1. Introduction

Two boundary questions relate the Oka property to Levi geometry. For a compact domain $K \subset \mathbb{C}^n$, $n > 1$, with smooth boundary, Forstnerič [9, Problem 4.22(a),(b)] asks whether an Oka complement forces K to be pseudoconvex, and conversely whether pseudoconvexity of K forces its complement to be Oka. The corresponding formulation for closed sets appears in Forstnerič [8, Problem 3.10(a),(b)]. Forstnerič and Wold [15, Problem 1.9] formulate the same two directions for smoothly bounded Oka domains and compact strongly pseudoconvex obstacles. For smooth compact domains in Euclidean space, the two implications have different dimension dependence. Pseudoconvexity is sufficient but not necessary when $n \geq 3$; in $\mathbb{C}^2$ it is necessary but not sufficient. We establish this distinction in general ambient manifolds. On Stein manifolds, the condition that the tangential Levi form have at least two nonnegative eigenvalues allows the Oka property of $M \setminus L$, where $L \supset K$ is compact, to be extended to $M \setminus K$. On arbitrary complex surfaces, a local construction gives a strongly pseudoconvex compact set K and a prescribed point $q \in M \setminus K$ such that every holomorphic map $f : \mathbb{C} \to M \setminus K$ with $f(0) = q$ is constant.

Throughout the paper, a compact set with smooth boundary is understood to be the closure of its interior. Near every $q \in \partial K$, an oriented defining function is a smooth real-valued function ρ on a neighbourhood of q such that

$$K = \{\rho \leq 0\}, \qquad d\rho \neq 0 \text{ on } \partial K. \tag{1.1}$$

Here and below, smooth means C^∞, and equalities such as $K = \{\rho \leq 0\}$ are local when ρ is defined only near a boundary point. In local holomorphic coordinates, the complex tangent space and the tangential Levi form are

$$T^{1,0}_q \partial K = \ker \partial \rho_q, \qquad \mathcal{L}_\rho(q; v) = \sum_{j,k=1}^{n} \rho_{z_j \bar{z}_k}(q) v_j \bar{v}_k \quad (v \in T^{1,0}_q \partial K).$$

The boundary is called pseudoconvex with respect to K when $\mathcal{L}_\rho(q; v) \geq 0$ for every $q \in \partial K$ and $v \in T^{1,0}_q \partial K$. With the boundary orientation induced by the open set $M \setminus K$, the defining function is $-\rho$, and the same hypersurface is pseudoconcave. It is *strongly pseudoconvex* with respect to K when this restricted Levi form is positive definite at every

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boundary point. For a real C^2 function, plurisubharmonicity and strict plurisubharmonicity mean that its Levi form is, respectively, positive semidefinite and positive definite. For continuous functions we use the equivalent restriction-to-complex-curves definition.

In Varolin's terminology [25], a complex manifold has the *density property* if the Lie algebra generated by its complete holomorphic vector fields is dense in the Lie algebra of all holomorphic vector fields for the compact-open topology; a vector field is complete when its holomorphic flow exists for every complex time. We use the convex approximation characterization of the Oka property: a complex manifold Y is Oka if, for every $m \geq 1$, every compact convex set $Q \subset \mathbb{C}^m$, and every holomorphic map to Y on a neighbourhood of Q, the map can be approximated uniformly on Q by entire maps $\mathbb{C}^m \to Y$ [5, Theorem 0.1]. For a compact set A in a Stein manifold M, write

$$\widehat{A}_{\mathcal{O}(M)} = \{x \in M : |f(x)| \leq \max_{A} |f| \text{ for every } f \in \mathcal{O}(M)\}.$$

The set A is $\mathcal{O}(M)$-*convex*, or holomorphically convex, when $\widehat{A}_{\mathcal{O}(M)} = A$. For $M = \mathbb{C}^n$, this is equivalent to polynomial convexity. A complex manifold M is Stein when global holomorphic functions separate points of M and $\widehat{A}_{\mathcal{O}(M)}$ is compact for every compact set $A \subset M$. A real-valued function is an exhaustion if each of its sublevel sets is compact.

Theorem 1.1. *Let M be a connected Stein manifold of complex dimension $n \geq 3$ with the density property. Let $K \Subset M$ have smooth boundary. Suppose that, for every $q \in \partial K$, the restriction of $\mathcal{L}_\rho(q;\cdot)$ to $T^{1,0}_q \partial K$ has at least two nonnegative eigenvalues, where ρ is any local defining function satisfying* (1.1)*. Then $M \setminus K$ is a connected Oka manifold.*

The eigenvalue condition concerns inertia and is independent of the Hermitian metric used to represent the Levi form. It is equivalent to having at most $n-3$ negative tangential eigenvalues. In dimension three it is precisely pseudoconvexity. In every dimension $n \geq 3$, it includes all smooth pseudoconvex boundaries, since all $n-1$ tangential eigenvalues are then nonnegative. Theorem 2.11 proves more generally that it suffices to have a compact set $L \supset K$ for which $M \setminus L$ is Oka. When M has the density property, such an L is obtained from Kusakabe's complement theorem.

Pseudoconvexity is not necessary for an Oka complement in any dimension $n \geq 3$. The following quartic models give explicit examples in Euclidean space.

Theorem 1.2. *For every integer $n \geq 3$, let*

$$K_n = \left\{(z,w) \in \mathbb{C}^{n-1} \times \mathbb{C} : |z|^2 - |w|^2 + |w|^4 \leq 1\right\}, \qquad X_n = \mathbb{C}^n \setminus K_n.$$

Then the following assertions hold.

(i) *K_n is the closure of a bounded star-shaped domain with smooth real-algebraic boundary. At $p_n = (1,0,\ldots,0;0) \in \partial K_n$, the tangential Levi form has $n-2$ positive eigenvalues and one negative eigenvalue. In particular, ∂K_n is not pseudoconvex.*

(ii) *X_n is a connected Oka manifold.*

Theorem 1.2 answers the necessity questions [9, Problem 4.22(a)] and [8, Problem 3.10(a)] negatively for every $n \geq 3$, even with a compact set and a smooth real-algebraic boundary. It also gives a negative answer in these dimensions to [15, Problem 1.9(b)]. In dimension two, a smoothly bounded Oka domain is pseudoconcave, as proved immediately after [15, Problem 1.9]. Indeed, the complex tangent space is one dimensional, so a negative Levi eigenvalue for a defining function ρ satisfying $K = \{\rho \leq 0\}$ would make the Levi form of $-\rho$ positive at the corresponding boundary point of $\mathbb{C}^2 \setminus K$. Such a strongly pseudoconvex boundary point produces a nonconstant bounded above plurisubharmonic function, which an Oka manifold cannot admit. Thus three is the smallest complex dimension for the counterexamples in Theorem 1.2.

Scaled copies of the quartic examples also occur in every prescribed nonempty open subset of a Stein manifold with the density property. More precisely, Theorem 3.3 constructs

there a compact set with connected interior and real-analytic boundary whose tangential Levi form has $n-2$ positive eigenvalues and one negative eigenvalue at one boundary point; its complement remains connected and Oka. The proof transfers the scaled set by a holomorphic coordinate map and applies Proposition 2.4 to explicit affine discs contained in that coordinate chart.

In complex dimension two, even real-analytic strong pseudoconvexity does not imply the Oka property of a compact complement. The counterexample is a sublevel set of a single explicit strictly plurisubharmonic rational expression.

Theorem 1.3. *For $T>0$, set $P(z,w)=z^5-w^2-1$ and consider the sublevel set*

$$K_T=\left\{(z,w)\in\mathbb{C}^2\setminus\{P=0\}:|z|^2+|w|^2+\frac{1}{|P(z,w)|^2}\le T\right\},\qquad p=(1,0). \tag{1.2}$$

There is a number $T_0>6$ such that, for every $T>T_0$:

(i) *K_T is the closure of a bounded connected domain with real-analytic strongly pseudoconvex boundary.*

(ii) *$X_T=\mathbb{C}^2\setminus K_T$ is connected, $p\in X_T$, and every holomorphic map $f:\mathbb{C}\to X_T$ with $f(0)=p$ is constant. In particular, X_T is not Oka.*

Theorem 1.3 gives a negative answer in dimension two to the compact-complement questions in [9, Problem 4.22(b)], [8, Problem 3.10(b)], and [15, Problem 1.9(a)].

The absence of nonconstant entire curves through the point p in Theorem 1.3(ii) can be reproduced on every complex surface. For every connected complex surface M, point $q\in M$, and open neighbourhood V of q, Theorem 5.2 constructs a compact set $K\Subset V$ that is the closure of a connected domain, has real-analytic strongly pseudoconvex boundary, and does not contain q, such that $M\setminus K$ is connected and every entire curve through q is constant. The construction applies to compact and noncompact complex surfaces. Corollary 5.4 also shows that, if $M\setminus K$ is Oka, the tangential Levi form of an oriented defining function for K cannot be negative definite at any boundary point. In dimension two this condition is precisely the pseudoconvexity of ∂K with respect to K.

Further results on compact sets that can be avoided by entire curves, the hull operations associated with this avoidance property, and decreasing families of Oka complements are developed in [4].

The historical route to these questions begins with the classical Oka principle. Oka [24] established it for the second Cousin problem, and Grauert [16] extended it to holomorphic bundles with complex Lie structure groups over Stein spaces. Gromov [17] introduced ellipticity: a holomorphic vector bundle $E\to Y$ admits a holomorphic map $s:E\to Y$ such that $s(0_y)=y$ and $d_{0_y}s(E_y)=T_yY$ for every $y\in Y$. This became a geometric source of the Oka principle. Forstnerič [5, 6] then characterized Oka manifolds by convex approximation and proved the equivalence of the approximation, interpolation, and parametric forms. The survey of Forstnerič and Lárusson [11] and the monograph [7] provide broader accounts of this development.

Compact complements emerged early as a test of how far this flexibility is controlled by the geometry of the omitted set. The complement of a closed Euclidean ball appears in Forstnerič [5, Problem 6.3]. Forstnerič and Ritter [13, Theorem 1] obtained approximation and interpolation for maps from Stein manifolds of dimension less than n into complements of compact convex subsets of $\mathbb{C}^n$. The decisive advance for compact omitted sets is Kusakabe's theorem: every compact holomorphically convex subset of a Stein manifold with the density property has Oka complement [21, Theorem 1.2]; in particular, every compact polynomially convex subset of $\mathbb{C}^n$, $n>1$, has Oka complement [21, Corollary 1.3]. Forstnerič and Wold [14] gave an alternative proof using holomorphic families of Fatou–Bieberbach domains, that is, proper subdomains of $\mathbb{C}^n$ biholomorphic to $\mathbb{C}^n$, and Kusakabe's elliptic characterization [20]. Their later work [15] treats geometric conditions for unbounded

convex complements and projective changes of affine chart. These results lead directly to the boundary-geometric problems in [9, Problem 4.22] and [15, Problem 1.9]. Theorem 1.1 replaces convexity of the compact set by a tangential Levi condition in dimensions at least three, including arbitrarily degenerate semipositive Levi forms.

Two adjacent results concern omitted sets of different types and do not settle the compact-domain boundary problem. Kusakabe [19] proved that a closed countable set $S \subset \mathbb{C}^n$, $n > 1$, with discrete set of limit points has Oka complement when it is tame, meaning that some $\Phi \in \mathrm{Aut}(\mathbb{C}^n)$ satisfies $\overline{\Phi(S)}^{\mathbb{P}^n} \not\supset \mathbb{P}^{n-1}_\infty$, where $\mathbb{P}^{n-1}_\infty = \mathbb{P}^n \setminus \mathbb{C}^n$ is the hyperplane at infinity. Forstnerič and Kusakabe [10] related the Oka property of tube domains in holomorphic line bundles to curvature positivity. The present paper instead concerns compact sets whose smooth real hypersurface boundaries satisfy specified inequalities for the Levi form.

The density property enters through approximation by holomorphic automorphisms. Andersén and Lempert [1] established approximation by holomorphic automorphisms of Euclidean space, and Varolin [25] formulated the density property for general complex manifolds. In the present proof, Kusakabe's theorem is applied to a compact $\mathcal{O}(M)$-convex set L to obtain that $M \setminus L$ is Oka; the passage from $M \setminus L$ to $M \setminus K$ is then proved by local arguments near ∂K. The fact that a connected Stein manifold of complex dimension at least two has at most one non-relatively-compact component after removal of a compact set [2, Section 5] is used in Lemma 2.10 to prove the required connectedness.

The positive result is proved by propagating the Oka property from the complement of a larger holomorphically convex compact set toward ∂K. The Levi hypothesis supplies holomorphic discs along which a suitable exhaustion function increases. If the propagation stopped, that function would attain a forbidden local maximum on the remaining set. In the quartic examples, the same local-to-global principle is driven instead by affine complex lines contained in the complement, allowing one negative tangential Levi direction without destroying the Oka property.

The surface counterexamples use a different idea. A nontrivial covering associated with an affine curve of genus two carries a bounded plurisubharmonic function whose differential detects motion transverse to the curve. Applied to the lift of an entire curve, this function and Liouville's theorem force the curve to be constant. Because the construction is local in the ambient surface, it can be placed in a coordinate neighbourhood of an arbitrary complex surface while the covering and the plurisubharmonic function are extended over the exterior region.

This use of a nontrivial fundamental group is specific to complex dimension two in the present construction. Zerón [26, Corollary 3.1] proved that the complement in $\mathbb{C}^n$ of a compact set with a Stein neighbourhood basis, meaning that every neighbourhood of the set contains a Stein neighbourhood, is $(n-2)$-connected. Such complements are simply connected for $n \geq 3$, whereas the universal covering associated with $\pi_1(\overline{C})$ used in Proposition 4.2 is nontrivial in dimension two.

Section 2 proves Proposition 2.6 and Theorem 2.11. Section 3 proves the properties of the sets K_n and transfers scaled copies of them to Stein manifolds. Section 4 constructs the bounded plurisubharmonic function on $\widetilde{Y}_R$ and proves Theorem 1.3. Section 5 constructs the analogous bounded plurisubharmonic function on a covering of an open subset of an arbitrary complex surface and proves Corollary 5.4.

2. Local approximation and complements of compact sets

For a compact subset A of a complex manifold, $\mathcal{O}(\mathrm{Op}\, A, Y)$ denotes holomorphic maps to Y on an open neighbourhood of A. The neighbourhood may depend on the map. The disc of radius $r > 0$ in $\mathbb{C}$ is Δ_r, and B_r denotes the Euclidean ball of radius r centred at the origin in the indicated real or complex Euclidean space.

A complex manifold Y has the *convex approximation property* if, for every integer $m \geq 1$, every compact convex set $Q \subset \mathbb{C}^m$, and every $f \in \mathcal{O}(\operatorname{Op} Q, Y)$, there are entire maps $F_\nu \colon \mathbb{C}^m \to Y$ such that

$$\sup_{z \in Q} d_Y\big(F_\nu(z), f(z)\big) \longrightarrow 0.$$

We use the characterization of Oka manifolds by the convex approximation property [5, Theorem 0.1], [6, Theorem 1.1 and Definition 1.2]. The distance d_Y is a complete distance inducing the topology of Y. On compact source sets, uniform approximation to a fixed map is independent of the compatible target distance. Euclidean distances used later for the geometry of ∂K are denoted by $|\cdot|$ and dist. Norms of linear maps are the operator norms induced by the Euclidean norms.

Lemma 2.1. *Let Z be a complex manifold of positive dimension and $q \in Z$. If every holomorphic map $f : \mathbb{C} \to Z$ with $f(0) = q$ is constant, then the connected component of q in Z is not Oka.*

Proof. If the component of q were Oka, Du–Guo–Wang–Xie [3, Theorem 1.3] would give a holomorphic map $H : \mathbb{C}^m \to Z$, where m is the dimension of that component, such that $H(0) = q$ and dH_0 has rank m. Restricting H to a complex line on which dH_0 is nonzero would give a nonconstant entire curve through q, contrary to the hypothesis. □

We recall the local approximation property of Du, Guo, Wang, and Xie [3, Definition 2.1]. A *special planar disc pair* $D \subset D'$ consists of compact topological discs in $\mathbb{C}$ with piecewise smooth boundaries, where D' is obtained by adjoining a disc to D along a proper boundary arc. For a connected complex manifold Y of dimension n, a point y belongs to $\mathcal{G}(Y)$ if there is an open neighbourhood V of y with the following property. For every such pair, every $m \geq 0$, closed polydiscs $P_0 \Subset \operatorname{int} P \subset \mathbb{C}^m$, and $f \in \mathcal{O}(\operatorname{Op}(D \times P), V)$, there are maps

$$F_\nu \in \mathcal{O}\big(\operatorname{Op}(D' \times P_0), Y\big), \qquad \sup_{D \times P_0} d_Y(F_\nu, f) \longrightarrow 0. \tag{2.1}$$

For $m = 0$, both parameter polydiscs are the one-point space $\mathbb{C}^0$. This is the local approximation property in [3, Definition 2.1]. All approximants in (2.1) take values in the ambient manifold Y.

Lemma 2.2. *For a connected complex manifold Y of dimension n, the set $\mathcal{G}(Y)$ is open. Every open submanifold $U \subset Y$ that is Oka is contained in $\mathcal{G}(Y)$.*

Proof. If $n = 0$, the connected manifold Y consists of one point, and both assertions follow from constant maps. Assume $n \geq 1$. If the neighbourhood V in the definition of $\mathcal{G}(Y)$ works for one point, the same V works for every point of V. Hence $\mathcal{G}(Y)$ is open.

Suppose that $U \subset Y$ is an open submanifold that is Oka, and fix $y \in U$. By Du–Guo–Wang–Xie [3, Theorem 1.3], there is a holomorphic map

$$H \colon \mathbb{C}^n \longrightarrow U, \qquad H(0) = y, \qquad \operatorname{rank} dH_0 = n.$$

Viewed as a map into Y, this map has full rank at 0. Du–Guo–Wang–Xie [3, Lemma 2.3] state that the existence of such a map implies the local approximation property used to define $\mathcal{G}(Y)$ at y. Hence $y \in \mathcal{G}(Y)$. □

We use the following two approximation results of Du, Guo, Wang, and Xie.

Theorem 2.3 ([3, Theorem 2.2])**.** *If $\mathcal{G}(Y) = Y$ for a connected complex manifold Y, then Y is Oka.*

Proposition 2.4 ([3, Proposition 5.3])**.** *Let $\vartheta \colon U \to \Omega \subset \mathbb{C}^n$ be a holomorphic coordinate chart of a connected complex manifold Y. Suppose that $x \in U$, $v \in \mathbb{C}^n \setminus \{0\}$, and $r > 0$ satisfy*

$$\vartheta(x) + \overline{\Delta}_r v \subset \Omega, \qquad \vartheta^{-1}\big(\vartheta(x) + \partial\Delta_r v\big) \subset \mathcal{G}(Y). \tag{2.2}$$

Then $x \in \mathcal{G}(Y)$.

The geometric argument uses the existence of at least two positive eigenvalues of a Levi form. Our convention for a real C^2 function u is

$$\mathcal{L}_u(q;v)=\sum_{j,k=1}^{n} u_{z_j\bar{z}_k}(q)v_j\bar{v}_k, \qquad \partial u_q(v)=\sum_{j=1}^{n} u_{z_j}(q)v_j.$$

The number of positive eigenvalues is independent of holomorphic coordinates, by the transformation rule for this Hermitian form.

Lemma 2.5. *Let u be a real C^2 function on a neighbourhood of $q\in\mathbb{C}^n$. If $\mathcal{L}_u(q;\cdot)$ has at least two positive eigenvalues, there are $v\neq 0$, $w\in\mathbb{C}^n$, and $r>0$ such that*

$$\gamma(\zeta)=q+\zeta v+\zeta^2 w, \qquad u(\gamma(\zeta))\geq u(q)+\frac{1}{2}\mathcal{L}_u(q;v)|\zeta|^2 \quad (0<|\zeta|\leq r), \tag{2.3}$$

with $\mathcal{L}_u(q;v)>0$. The closed disc $\gamma(\overline{\Delta}_r)$ is embedded and lies in a single holomorphic coordinate chart in which it is a complex-line disc. The radius r may be decreased arbitrarily.

Proof. There is a two-dimensional complex subspace $E\subset\mathbb{C}^n$ on which $\mathcal{L}_u(q;\cdot)$ is positive definite. The complex linear form ℓ and homogeneous quadratic form Q given by

$$\ell(h)=\sum_{j=1}^{n} u_{z_j}(q)h_j, \qquad Q(h)=\sum_{j,k=1}^{n} u_{z_jz_k}(q)h_jh_k$$

give the second-order Taylor expansion

$$u(q+h)=u(q)+2\operatorname{Re}\ell(h)+\mathcal{L}_u(q;h)+\operatorname{Re}Q(h)+o(|h|^2). \tag{2.4}$$

If $\ell\neq 0$, then $\dim_{\mathbb{C}}(E\cap\ker\ell)\geq 1$. Choose $v\in E\cap\ker\ell$ with $v\neq 0$. Surjectivity of $\ell\colon\mathbb{C}^n\to\mathbb{C}$ gives a vector w satisfying

$$2\ell(w)+Q(v)=0. \tag{2.5}$$

For $h=\zeta v+\zeta^2 w$, the linear term in ζ vanishes, and (2.5) cancels the harmonic quadratic term. More explicitly,

$$2\operatorname{Re}\ell(h)+\operatorname{Re}Q(h)=\operatorname{Re}\big((2\ell(w)+Q(v))\zeta^2\big)+O(|\zeta|^3)=O(|\zeta|^3),$$
$$\mathcal{L}_u(q;h)=\mathcal{L}_u(q;v)|\zeta|^2+O(|\zeta|^3).$$

Substitution into (2.4) yields

$$u(q+\zeta v+\zeta^2 w)=u(q)+\mathcal{L}_u(q;v)|\zeta|^2+o(|\zeta|^2). \tag{2.6}$$

If $\ell=0$, the restriction $Q|_E$ has a nonzero zero. To see this explicitly, take a basis e_1,e_2 of E. If $Q(e_2)=0$, take $v=e_2$. Otherwise the polynomial $t\mapsto Q(e_1+te_2)$ has degree two and hence has a complex root t_0; take $v=e_1+t_0e_2$. In both cases $v\neq 0$, $Q(v)=0$, and $\mathcal{L}_u(q;v)>0$. With $w=0$, formula (2.4) again gives (2.6).

In either case, $\mathcal{L}_u(q;v)>0$, so the remainder in (2.6) has absolute value at most $\frac{1}{2}\mathcal{L}_u(q;v)|\zeta|^2$ for all sufficiently small nonzero ζ. This proves (2.3).

Complete v to a complex basis $v,b_2,\dots,b_n$ of $\mathbb{C}^n$. The holomorphic map

$$\Psi(\zeta,t_2,\dots,t_n)=q+\zeta v+\zeta^2 w+\sum_{j=2}^{n} t_jb_j$$

has invertible differential at the origin, with columns $v,b_2,\dots,b_n$. The holomorphic inverse function theorem gives an open neighbourhood W of the origin on which Ψ is biholomorphic. After restriction, its image is contained in the domain of u. Decrease r so that

$$\overline{\Delta}_r\times\{0\}^{n-1}\Subset W.$$

Then $\vartheta=\Psi^{-1}$ is a single chart containing the whole closed disc and

$$\vartheta(\gamma(\zeta))=(\zeta,0,\dots,0) \qquad (|\zeta|\leq r).$$

In particular, the disc is embedded. □

Proposition 2.6. *Let Y be a connected complex manifold, with $S = Y \setminus \mathcal{G}(Y)$. Suppose that u is a real C^2 function on a neighbourhood of $q \in S$ and that $u|_S$ has a local maximum at q. Then $\mathcal{L}_u(q;\cdot)$ has at most one positive eigenvalue.*

Proof. A neighbourhood U_0 of q satisfies

$$u(s) \le u(q) \qquad (s \in S \cap U_0).$$

If the Levi form has two positive eigenvalues, apply Lemma 2.5 in coordinates centred at q, with the whole closed disc contained in U_0. Its boundary satisfies

$$u(\gamma(\zeta)) > u(q) \qquad (|\zeta| = r),$$

and hence $\gamma(\partial\Delta_r) \subset \mathcal{G}(Y)$. The coordinate chart supplied by Lemma 2.5 places this disc in the form (2.2). Proposition 2.4 therefore gives $q \in \mathcal{G}(Y)$, contrary to $q \in S$. □

We next construct a function on $M \setminus K$ whose Levi form has a prescribed number of positive eigenvalues and which tends to $-\infty$ at ∂K. Let M be a Stein manifold of complex dimension $n \ge 2$ and $K \Subset M$ a compact set with smooth boundary, with the orientation specified in (1.1). The inertia of the tangential Levi form is independent of the oriented defining function and the Hermitian metric. If r and r' are two oriented defining functions on a common neighbourhood of a boundary point, Hadamard's lemma gives $r' = ar$, where a is smooth and positive after the neighbourhood is decreased. Consequently,

$$\mathcal{L}_{r'}(q;v) = a(q)\mathcal{L}_r(q;v) \qquad (q \in \partial K,\ v \in \ker \partial r_q).$$

The terms containing r or $\partial r(v)$ vanish. A change of Hermitian metric changes the representing matrix by a congruence, which preserves the numbers of positive, negative, and zero eigenvalues.

Lemma 2.7. *Suppose that $1 \le k \le n-1$ and that the tangential Levi form of ∂K has at least k nonnegative eigenvalues at every point. For every smooth Hermitian metric h on M, there are numbers $\delta > 0$ and $C_0 \ge 0$, a smooth oriented defining function r on a neighbourhood of $\Sigma = \partial K$, and a collar parametrization*

$$E : \Sigma \times (-2\delta, 2\delta) \longrightarrow M, \qquad r(E(q,t)) = t,$$

with the following property. For each $x = E(q,t)$ with $0 < t < \delta$, there is a complex k-plane $V_x \subset \ker \partial r_x$ such that

$$\mathcal{L}_r(x;v) \ge -C_0 r(x) h_x(v,v) \qquad (v \in V_x).$$

The collar has compact closure in M.

Proof. The assertion is vacuous when $K = \varnothing$; assume that $K \ne \varnothing$. Smooth local oriented defining functions r_α on a finite neighbourhood cover of Σ, together with a subordinate smooth partition of unity η_α, give

$$r = \sum_\alpha \eta_\alpha r_\alpha.$$

On the intersection of their domains with $\operatorname{int} K$, all r_α are negative; on the other side of Σ, they are all positive. On Σ,

$$dr = \sum_\alpha \eta_\alpha \, dr_\alpha \ne 0,$$

because the nonzero covectors dr_α are positive multiples of the same oriented conormal. After restriction to a smaller neighbourhood of the compact hypersurface Σ, the function r is smooth, dr is nowhere zero, and

$$K \cap \operatorname{dom} r = \{r \le 0\}.$$

For a smooth Riemannian metric g on M, the real vector field

$$N = \frac{\nabla_g r}{|\nabla_g r|_g^2}$$

satisfies $dr(N) = 1$. Its flow Φ_t is available for $q \in \Sigma$ and $|t| < a$, with one $a > 0$, by compactness of Σ. Along each such flow line,

$$\frac{d}{dt} r(\Phi_t(q)) = 1, \qquad r(\Phi_t(q)) = t. \tag{2.7}$$

The map $E(q,t) = \Phi_t(q)$ has invertible differential at $(q,0)$, since $N(q)$ is transverse to $T_q\Sigma$. It is injective: if $E(q,t) = E(q',t')$, then (2.7) gives $t = t'$, and uniqueness of the flow gives $q = q'$ after application of Φ_{-t}. Moreover,

$$dE_{(q,t)}(\xi, s) = d\Phi_t(q)\big(\xi + sN(q)\big), \qquad \xi \in T_q\Sigma,$$

is invertible. The inverse function theorem therefore makes E a diffeomorphism onto an open neighbourhood of Σ. A number $\delta > 0$ with $2\delta < a$ makes $E(\Sigma \times [-2\delta, 2\delta])$ compact.

The complex hyperplanes

$$H_x = \ker \partial r_x$$

form a smooth complex vector bundle of rank $n-1$ over the collar. To see its local frames explicitly, choose holomorphic coordinates in which $r_{z_n} \neq 0$. The vectors

$$e_j = \frac{\partial}{\partial z_j} - \frac{r_{z_j}}{r_{z_n}} \frac{\partial}{\partial z_n}, \qquad 1 \le j \le n-1,$$

form a smooth frame of H. Smooth Gram–Schmidt orthonormalization with respect to h gives an h-orthonormal frame $e_1', \dots, e_{n-1}'$ on a smaller neighbourhood.

Choose finitely many relatively compact coordinate neighbourhoods $W_\alpha \subset \Sigma$ whose closures cover Σ and on whose collars such orthonormal frames exist. Decreasing δ keeps all the products $\overline{W}_\alpha \times [-\delta, \delta]$ in their respective frame domains. The matrices

$$B_\alpha(q,t)_{ij} = \mathcal{L}_r\big(E(q,t); e_j'(E(q,t)), e_i'(E(q,t))\big)$$

are Hermitian and smooth in (q,t). Here the two-vector notation denotes the sesquilinear polarization of the Levi form. Compactness gives finite constants

$$C_\alpha = \sup_{\substack{q \in \overline{W}_\alpha \\ |t| \le \delta}} \left\| \frac{\partial B_\alpha}{\partial t}(q,t) \right\|, \qquad C_0 = \max_\alpha C_\alpha.$$

The fundamental theorem of calculus yields

$$\|B_\alpha(q,t) - B_\alpha(q,0)\| \le C_0 |t|.$$

Fix $x = E(q,t)$ with $0 < t < \delta$, and select an index α with $q \in \overline{W}_\alpha$. By the spectral theorem and the hypothesis on the number of nonnegative eigenvalues, the span $A_q \subset \mathbb{C}^{n-1}$ of k eigenvectors of $B_\alpha(q,0)$ with nonnegative eigenvalues satisfies

$$\xi^* B_\alpha(q,0) \xi \ge 0 \qquad (\xi \in A_q).$$

For $v = \sum_i \xi_i e_i'(x)$ with $\xi \in A_q$,

$$\mathcal{L}_r(x; v) = \xi^* B_\alpha(q,t) \xi \ge -C_0 t |\xi|^2 = -C_0 r(x) h_x(v,v). \tag{2.8}$$

Let $V_x = \{\sum_i \xi_i e_i'(x) : \xi \in A_q\}$. Then $\dim_{\mathbb{C}} V_x = k$, and (2.8) is the estimate in the statement of the lemma. □

Proposition 2.8. *Let M be a Stein manifold of complex dimension $n \ge 2$ and $K \Subset M$ a compact set with smooth boundary. Suppose that its tangential Levi form has at least k nonnegative eigenvalues everywhere, where $1 \le k \le n-1$. There is a smooth real function F on $X = M \setminus K$ such that:*

(i) *$\mathcal{L}_F$ has at least k positive eigenvalues at every point of X;*

(ii) *for every $c \in \mathbb{R}$, there is a neighbourhood U_c of ∂K in M such that $F < c$ on $U_c \setminus K$.*

Equivalently, $F(x) \to -\infty$ uniformly as x approaches ∂K from $M \setminus K$.

Proof. A proper holomorphic embedding $H : M \hookrightarrow \mathbb{C}^N$ exists by the Stein embedding theorem; see [8, Section 2]. The function

$$\tau(x) = \|H(x)\|^2$$

is a smooth proper exhaustion. In fact, $\{\tau \le c\} = H^{-1}(\overline{B}_{\sqrt{c}})$ is compact for $c \ge 0$, and the sublevel set is empty for $c < 0$. For every $0 \ne v \in T^{1,0}_x M$,

$$\mathcal{L}_\tau(x; v) = \|dH_x(v)\|^2 > 0,$$

because H is an immersion. Its Levi form gives the Hermitian metric

$$h_x(v, w) = \mathcal{L}_\tau(x; v, w). \tag{2.9}$$

For $K = \varnothing$, the choice $F = \tau$ proves the assertion. Assume henceforth that $K \ne \varnothing$, and take r, E, δ, C_0 from Lemma 2.7 for this metric. A further decrease makes $0 < \delta < 1$.

Choose a smooth function $\chi : \mathbb{R} \to [0, 1]$ such that

$$\chi(t) = 1 \quad (0 \le t \le \delta/3), \qquad \chi(t) = 0 \quad (t \ge 2\delta/3).$$

On X, the function

$$\psi(x) = \begin{cases} \chi(r(x)) \log r(x), & x \in E(\Sigma \times (0, \delta)), \\ 0, & x \notin E(\Sigma \times (0, \delta)) \end{cases}$$

is smooth, because the first expression vanishes on the entire band $2\delta/3 < r < \delta$ before the collar ends. In the inner collar $0 < r < \delta/3$,

$$\mathcal{L}_\psi(x; v) = \frac{\mathcal{L}_r(x; v)}{r(x)} - \frac{|\partial r_x(v)|^2}{r(x)^2}.$$

For the k-plane $V_x \subset \ker \partial r_x$ supplied by Lemma 2.7, this gives

$$\mathcal{L}_\psi(x; v) \ge -C_0 h_x(v, v) \qquad (v \in V_x). \tag{2.10}$$

The transition band

$$T = E\big(\Sigma \times [\delta/3, 2\delta/3]\big)$$

is compact. The h-unit sphere bundle over T is compact, so continuity of the Levi form gives $C_1 \ge 0$ with

$$\mathcal{L}_\psi(x; v) \ge -C_1 h_x(v, v) \qquad (x \in T,\ v \in T^{1,0}_x M). \tag{2.11}$$

For any constant $A > \max\{C_0, C_1\}$, consider

$$F = A\tau + \psi.$$

(i) On the inner collar, (2.9) and (2.10) yield

$$\mathcal{L}_F(x; v) \ge (A - C_0) h_x(v, v) > 0 \qquad (0 \ne v \in V_x).$$

A Hermitian form positive definite on a k-plane has at least k positive eigenvalues: otherwise its nonpositive spectral subspace has dimension greater than $n - k$ and meets that plane nontrivially. On T, (2.11) gives

$$\mathcal{L}_F(x; v) \ge (A - C_1) h_x(v, v) > 0 \qquad (v \ne 0).$$

Outside the inner collar and T, the function ψ vanishes locally, and $\mathcal{L}_F = Ah$ is positive definite.

(ii) The number

$$M_0 = \max_{E(\Sigma \times [0, \delta])} A\tau$$

is finite. On $0 < r < \delta/3$,

$$F(x) = A\tau(x) + \log r(x) \le M_0 + \log r(x).$$

For $c \in \mathbb{R}$, take

$$0 < \varepsilon < \min\{\delta/3, e^{c-M_0}\}, \qquad U_c = E(\Sigma \times (-\varepsilon, \varepsilon)).$$

Then $F < c$ on $U_c \setminus K$. □

We next combine Proposition 2.8 with connectedness and with the local approximation property on $M \setminus L$.

Lemma 2.9. *Let M be a connected Stein manifold of complex dimension at least two. For every compact set $C \subset M$, the complement $M \setminus C$ has at most one connected component that is not relatively compact in M.*

Proof. This is the one-end theorem for Stein manifolds of complex dimension at least two; see [2, Section 5]. The conclusion is stated here in the component form needed below. □

Lemma 2.10. *Let M be a connected Stein manifold of complex dimension $n \ge 2$ and $K \Subset M$ a compact set with smooth boundary. If the tangential Levi form of ∂K has at least one nonnegative eigenvalue everywhere, then $M \setminus K$ is connected.*

Proof. For $K = \varnothing$, the assertion is the connectedness of M. Otherwise, Proposition 2.8, with $k = 1$, gives a function F on $X = M \setminus K$ with at least one positive Levi eigenvalue everywhere and with uniform decay at ∂K.

Suppose that a component U of X is relatively compact in M. Its boundary lies in ∂K. Indeed, a point of $\operatorname{int} K$ has a neighbourhood disjoint from U. If a boundary point belonged to X, a connected coordinate ball around it, contained in X and meeting U, would lie entirely in U, contradicting that the point belongs to ∂U.

For $x_0 \in U$, the set

$$A_0 = \{x \in U : F(x) \ge F(x_0)\}$$

is nonempty and compact in U. To check compactness, a sequence in A_0 has a subsequence converging in the compact set $\overline{U}$. Property (ii) of Proposition 2.8, applied with $c = F(x_0)$, excludes limits in ∂U. Hence the subsequential limit lies in U and belongs to A_0 by continuity. A maximum of F on A_0 is therefore a maximum on U. Let ϕ be a holomorphic coordinate chart centred at a maximum x_*. For every $v \in \mathbb{C}^n \cong T^{1,0}_{x_*}M$,

$$\mathcal{L}_F(x_*; v) = \frac{1}{4}\, \Delta_\zeta (F \circ \phi^{-1})(\zeta v)\big|_{\zeta=0} \le 0.$$

This contradicts the positive Levi eigenvalue of F. Consequently no component of X is relatively compact. By Lemma 2.9, at most one component remains. The set X is nonempty because a positive-dimensional Stein manifold is noncompact and K is compact. Hence X is connected. □

Theorem 2.11. *Let M be a connected Stein manifold of complex dimension $n \ge 3$ and $K \Subset M$ a compact set with smooth boundary. Suppose that the tangential Levi form of ∂K has at least two nonnegative eigenvalues at every boundary point. If a compact set $L \subset M$ contains K and $M \setminus L$ is Oka, then $M \setminus K$ is a connected Oka manifold.*

Proof. Lemma 2.10 gives connectedness of $X = M \setminus K$. Proposition 2.8, with $k = 2$, gives a smooth function F on X whose Levi form has at least two positive eigenvalues everywhere and which tends uniformly to $-\infty$ at ∂K.

The set $S = X \setminus \mathcal{G}(X)$ is closed in X, by Lemma 2.2. The open submanifold $M \setminus L$ is Oka and hence is contained in $\mathcal{G}(X)$ by the second assertion of Lemma 2.2. Hence

$$S \subset L \setminus K. \tag{2.12}$$

If $S \neq \varnothing$, take $x_0 \in S$ and consider

$$S_0 = \{x \in S : F(x) \geq F(x_0)\}.$$

The set S_0 is nonempty and compact in X. Indeed, every sequence in S_0 has a subsequence converging to a point of L by (2.12). Its limit cannot lie in $\operatorname{int} K$, since all sequence terms lie in X. It cannot lie in ∂K, by Proposition 2.8(ii) with $c = F(x_0)$. Thus the limit belongs to X, and the relative closedness of S and continuity of F place it in S_0.

The function F attains a maximum at some $x_* \in S_0$. Every point of $S \setminus S_0$ has a smaller value than $F(x_0)$, so x_* is also a maximum point of $F|_S$. Proposition 2.6 says that $\mathcal{L}_F(x_*; \cdot)$ has at most one positive eigenvalue, contrary to the construction of F. Therefore $S = \varnothing$. Theorem 2.3 yields the Oka property of X. □

Proof of Theorem 1.1. Choose a nonempty compact neighbourhood C of K in M. Its $\mathcal{O}(M)$-hull

$$L = \widehat{C}_{\mathcal{O}(M)} = \{x \in M : |f(x)| \leq \max_C |f| \text{ for every } f \in \mathcal{O}(M)\}$$

is compact by the holomorphic convexity of the Stein manifold M. Since $C \subset L$, $K \subset \operatorname{int} L$. Moreover, for every $f \in \mathcal{O}(M)$,

$$\max_L |f| = \max_C |f|, \qquad \widehat{L}_{\mathcal{O}(M)} = L.$$

Kusakabe's theorem [21, Theorem 1.2] therefore gives the Oka property of $M \setminus L$. Theorem 2.11 applies. □

For a pseudoconvex boundary, all $n-1$ tangential Levi eigenvalues are nonnegative. Thus Theorem 1.1 includes every compact set with smooth pseudoconvex boundary when $n \geq 3$. For $n = 3$, the condition of two nonnegative eigenvalues is precisely pseudoconvexity. For $n \geq 4$, it permits as many as $n-3$ negative tangential Levi eigenvalues at a boundary point.

Corollary 2.12. *For $n \geq 3$, every compact set in $\mathbb{C}^n$ with smooth pseudoconvex boundary has connected Oka complement.*

Proof. The space $\mathbb{C}^n$ has the density property by Andersén–Lempert theory [1]. All $n-1 \geq 2$ tangential Levi eigenvalues of ∂K are nonnegative, so Theorem 1.1 applies. □

3. Oka complements with mixed Levi signature

Proposition 2.4 also applies when every point of a compact complement lies on a complex affine line contained in the complement.

Proposition 3.1. *Let $n \geq 2$ and $L \subset \mathbb{C}^n$ be compact. Suppose that, for every $x \in \mathbb{C}^n \setminus L$, there exists $v_x \in \mathbb{C}^n \setminus \{0\}$ such that*

$$x + \mathbb{C}v_x \subset \mathbb{C}^n \setminus L. \tag{3.1}$$

Then $\mathbb{C}^n \setminus L$ is connected and Oka.

Proof. Compactness gives $R > 0$ with $L \subset B_R$. In this proof, $Y = \mathbb{C}^n \setminus L$ and $E = \mathbb{C}^n \setminus \overline{B}_R$. The set E is path connected. For every $x \in Y$, the path $t \mapsto x + tv_x$, $t \geq 0$, lies in Y by (3.1) and reaches E, since

$$|x + tv_x| \geq t|v_x| - |x| \longrightarrow \infty.$$

Thus every point of Y can be joined to E, and Y is path connected.

Kusakabe's theorem [21, Corollary 1.3] shows that E is Oka. It is an open submanifold of Y, so Lemma 2.2 yields $E \subset \mathcal{G}(Y)$. For $x \in Y$, a radius

$$r > \frac{R + |x|}{|v_x|}$$

satisfies

$$x + \overline{\Delta}_r v_x \Subset Y, \qquad |x + \zeta v_x| \geq r|v_x| - |x| > R \quad (|\zeta| = r). \tag{3.2}$$

The first inclusion follows because the closed disc is a compact subset of the affine line in (3.1). The second inequality places its boundary in $E \subset \mathcal{G}(Y)$. In the global coordinate chart $\mathrm{id}_Y \colon Y \to Y \subset \mathbb{C}^n$, the disc (3.2) therefore satisfies both conditions of Proposition 2.4. Hence $x \in \mathcal{G}(Y)$. Since x was arbitrary, $\mathcal{G}(Y) = Y$, and Theorem 2.3 proves that Y is Oka. □

Proof of Theorem 1.2. (i) On $\mathbb{C}^{n-1} \times \mathbb{C}$, write

$$\rho_n(z, w) = |z|^2 - |w|^2 + |w|^4 - 1, \qquad Z = |z|^2, \qquad W = |w|^2.$$

The inequality $\rho_n \leq 0$ gives

$$|(z, w)|^2 = Z + W \leq 1 + 2W - W^2 = 2 - (W - 1)^2 \leq 2.$$

Since K_n is closed, this proves compactness and $K_n \subset B_2$. The identity $\rho_n(0, 0) = -1$ shows that K_n has nonempty interior.

The complex first derivatives are

$$\frac{\partial \rho_n}{\partial z_j} = \bar{z}_j \quad (1 \leq j \leq n-1), \qquad \frac{\partial \rho_n}{\partial w} = (-1 + 2W)\bar{w}.$$

At a critical point of ρ_n, these equations imply $z = 0$ and either $W = 0$ or $W = \frac{1}{2}$. The corresponding values of ρ_n are -1 and $-\frac{5}{4}$. Therefore 0 is a regular value. For $q \in \{\rho_n = 0\}$ and real $s \to 0$, Taylor's formula gives

$$\rho_n\big(q + s\nabla\rho_n(q)\big) = s|\nabla\rho_n(q)|^2 + O(s^2).$$

Thus every neighbourhood of q contains points of both signs. Continuity of ρ_n then gives

$$\partial K_n = \{\rho_n = 0\}, \qquad \operatorname{int} K_n = \{\rho_n < 0\}.$$

The implicit function theorem makes the boundary smooth, and its polynomial equation makes it real algebraic. For $y = (z, w) \in K_n$ and $0 \leq t < 1$,

$$\rho_n(ty) = t^2\big(\rho_n(y) + 1\big) - t^2(1 - t^2)|w|^4 - 1 \leq t^2 - 1 < 0. \tag{3.3}$$

Consequently $\{\rho_n < 0\}$ is star-shaped about the origin and is connected. Letting t increase to 1 in (3.3) gives $K_n = \overline{\{\rho_n < 0\}}$.

For $(v, \eta) \in \mathbb{C}^{n-1} \times \mathbb{C}$, the full Levi form is

$$\mathcal{L}_{\rho_n}\big((z, w); (v, \eta)\big) = \sum_{j=1}^{n-1} |v_j|^2 + (-1 + 4|w|^2)|\eta|^2.$$

At $p_n = (1, 0, \ldots, 0; 0)$, one has $\rho_n(p_n) = 0$ and $\partial\rho_n(p_n) = dz_1$. Hence

$$T^{1,0}_{p_n}\partial K_n = \{(v, \eta) \in \mathbb{C}^{n-1} \times \mathbb{C} : v_1 = 0\},$$

$$\mathcal{L}_{\rho_n}\big(p_n; (v, \eta)\big) = \sum_{j=2}^{n-1} |v_j|^2 - |\eta|^2 \quad (v_1 = 0).$$

The restricted form has eigenvalue 1 with multiplicity $n - 2$ and eigenvalue -1 with multiplicity 1. With the orientation $K_n = \{\rho_n \leq 0\}$, this proves the failure of pseudoconvexity.

(ii) Fix $x = (z, w) \in X_n$. Since $n - 1 \geq 2$, choose a unit vector $v \in \mathbb{C}^{n-1}$ orthogonal to z. For every $\zeta \in \mathbb{C}$,

$$|z + \zeta v|^2 = |z|^2 + |\zeta|^2, \qquad \rho_n(z + \zeta v, w) = \rho_n(x) + |\zeta|^2 > 0.$$

Thus $x + \mathbb{C}(v, 0)$ is an affine complex line contained in X_n. Proposition 3.1, with $L = K_n$, proves that X_n is connected and Oka. □

The local quartic construction requires an Oka complement outside a small holomorphically convex neighbourhood. Such neighbourhoods can be chosen inside any prescribed coordinate chart.

Lemma 3.2. *Let M be a Stein manifold, $p \in M$, and $W \subset M$ an open neighbourhood of p. There is a compact $\mathcal{O}(M)$-convex set L with $p \in \operatorname{int} L$ and $L \subset W$.*

Proof. Choose a relatively compact open neighbourhood U with $p \in U \Subset W$. Since $\{p\}$ is $\mathcal{O}(M)$-convex, Forstnerič [7, Proposition 2.5.1] gives a smooth strongly plurisubharmonic function $\varphi : M \to \mathbb{R}$ such that

$$\varphi(p) < 0, \qquad \varphi > 1 \quad \text{on } M \setminus U.$$

Choose c with $\varphi(p) < c < 0$ and let $L = \{\varphi \le c\}$. Then $p \in \operatorname{int} L$ and $L \Subset U \Subset W$. For every $x \in M \setminus L$, the plurisubharmonic function φ satisfies $\varphi(x) > c \ge \max_L \varphi$. Hence the plurisubharmonic hull $\{x : u(x) \le \max_L u \text{ for every plurisubharmonic } u \text{ on } M\}$ equals L. On a Stein manifold the plurisubharmonic and holomorphic hulls agree [7, Corollary 2.5.3]; hence L is $\mathcal{O}(M)$-convex. □

Theorem 3.3. *Let M be a connected Stein manifold of complex dimension $n \ge 3$. Suppose that $M \setminus L$ is Oka for every compact $\mathcal{O}(M)$-convex set $L \subset M$. Every nonempty open set $V \subset M$ contains a compact set $K \Subset V$ that is the closure of a connected domain and has the following properties:*

(i) *∂K is real analytic, its tangential Levi form has at least $n-2$ positive eigenvalues everywhere, and at some point of ∂K it has exactly $n-2$ positive eigenvalues and one negative eigenvalue.*
(ii) *$M \setminus K$ is a connected Oka manifold.*

In particular, the conclusion holds for every connected Stein manifold with the density property and complex dimension at least three.

Proof. Choose a holomorphic coordinate embedding $j : B_R \to M$, with $j(B_R) \subset V$, and let $p_0 = j(0)$. Lemma 3.2 provides a compact $\mathcal{O}(M)$-convex neighbourhood L of p_0 such that

$$L \subset j(B_{R/4}). \tag{3.4}$$

For the compact quartic domain K_n of Theorem 1.2, a sufficiently small $\varepsilon > 0$ satisfies

$$\varepsilon K_n \subset j^{-1}(\operatorname{int} L).$$

Set $K = j(\varepsilon K_n)$.

(i) Theorem 1.2(i) gives compactness, connectedness, a real-analytic boundary, and exactly $n-2$ positive tangential Levi eigenvalues and one negative tangential Levi eigenvalue at $j(\varepsilon p_n)$. The number of positive eigenvalues at the other boundary points is verified as follows. In the coordinates $(z, w) \in \mathbb{C}^{n-1} \times \mathbb{C}$, the set K has the form $\{\rho_\varepsilon \le 0\}$, where

$$\rho_\varepsilon(z, w) = \frac{|z|^2 - |w|^2}{\varepsilon^2} + \frac{|w|^4}{\varepsilon^4} - 1.$$

Its Levi form restricts to $|v|^2/\varepsilon^2$ on $\mathbb{C}^{n-1} \times \{0\}$. The intersection of that subspace with $\ker \partial\rho_\varepsilon$ has complex dimension at least $n-2$. Hence the tangential Levi form has at least $n-2$ positive eigenvalues at every boundary point. Holomorphic coordinate changes preserve these statements.

(ii) Since $n-2 \ge 1$, Lemma 2.10 shows that $X = M \setminus K$ is connected. By the hypothesis on M and Lemma 2.2,

$$M \setminus L \subset \mathcal{G}(X). \tag{3.5}$$

For $x \in L \setminus K$, write $j^{-1}(x) = (z, w)$. There is a unit vector $v \in \mathbb{C}^{n-1}$ with $\langle v, z\rangle = 0$, since $n - 1 \ge 2$. Along the affine disc

$$\gamma_x(\zeta) = (z + \zeta v, w), \qquad |\zeta| \le R/2,$$

the identity

$$\rho_\varepsilon(\gamma_x(\zeta)) = \rho_\varepsilon(z,w) + \frac{|\zeta|^2}{\varepsilon^2} > 0$$

holds. Moreover, (3.4) gives

$$|\gamma_x(\zeta)| \le |(z,w)| + |\zeta| < \frac{3R}{4} < R, \qquad |\gamma_x(\zeta)|^2 = |(z,w)|^2 + \frac{R^2}{4} \quad (|\zeta| = R/2).$$

The closed disc therefore lies in $B_R \setminus \varepsilon K_n$, and its boundary lies outside $j^{-1}(L)$. In the coordinate chart $j^{-1} : j(B_R) \cap X \to B_R \setminus \varepsilon K_n$, Proposition 2.4, together with (3.5), yields $x \in \mathcal{G}(X)$. Every point of X belongs either to $M \setminus L$ or to $L \setminus K$, so $\mathcal{G}(X) = X$. Theorem 2.3 now gives the Oka property. If M has the density property, Kusakabe's complement theorem [21, Theorem 1.2] says precisely that $M \setminus L$ is Oka for every compact $\mathcal{O}(M)$-convex set L. □

4. A covering-space obstruction and explicit compact sets

The unit disc is denoted by $\Delta = \Delta_1$. Its pseudohyperbolic distance and positive Green function are

$$\rho_\Delta(\zeta, a) = \left|\frac{\zeta - a}{1 - \bar{a}\zeta}\right|, \qquad G(\zeta, a) = -\log \rho_\Delta(\zeta, a).$$

The Euclidean Laplacian has the normalization $-\Delta_\zeta G(\zeta, a) = 2\pi\delta_a$, where δ_a is the Dirac point mass at a. For $a \in \Delta$ and $0 < r < 1$, the notation $D_{\mathrm{ph}}(a, r)$ refers to the pseudohyperbolic disc $\{\rho_\Delta(\zeta, a) < r\}$.

Lemma 4.1. *Let $A \subset \Delta$ satisfy $0 \notin A$ and*

$$\inf\{\rho_\Delta(a,b) : a, b \in A,\ a \ne b\} > 0.$$

There are $\sigma > 0$ and a bounded continuous subharmonic function $u : \Delta \to \mathbb{R}$ such that u is constant on each $D_{\mathrm{ph}}(a, \sigma)$, $a \in A$, and u is harmonic on a neighbourhood of 0 with $du(0) \ne 0$.

Proof. If $A = \varnothing$, the function $u(\zeta) = \operatorname{Re}\zeta$ has the required properties. Assume $A \ne \varnothing$. The quantities

$$h(\zeta) = 1 - |\zeta|^2, \qquad h_a = 1 - |a|^2$$

will measure Euclidean distances to the boundary.

Step 1: an estimate for a Green potential. Direct differentiation gives

$$-\Delta\sqrt{h(\zeta)} = \frac{2 - |\zeta|^2}{h(\zeta)^{3/2}} \ge h(\zeta)^{-3/2}.$$

For $|\zeta| < r < 1$, Green's representation formula on Δ_r reads

$$\sqrt{h(\zeta)} = \sqrt{1 - r^2} + \frac{1}{2\pi}\int_{\Delta_r} G_r(\zeta, \eta)\frac{2 - |\eta|^2}{h(\eta)^{3/2}}\, dA(\eta),$$

where G_r is the positive Green function of Δ_r and dA is Euclidean area. The kernels G_r, extended by zero outside Δ_r, increase to G as $r \uparrow 1$. The integrands are nonnegative and the boundary term tends to zero. Monotone convergence therefore yields

$$\int_\Delta G(\zeta, \eta) h(\eta)^{-3/2}\, dA(\eta) \le 2\pi\sqrt{h(\zeta)}.$$

Choose $s > 0$ sufficiently small that the Euclidean discs

$$B(a, 10sh_a), \qquad a \in A, \tag{4.1}$$

are contained in Δ and are pairwise disjoint, and that

$$\frac{1}{2}h_a \le h(\eta) \le 2h_a \qquad \big(\eta \in B(a, 10sh_a)\big). \tag{4.2}$$

To verify that one number s can satisfy these requirements, note that the identity $h_a = (1-|a|)(1+|a|)$ gives containment when $20s < 1$. Moreover, $|h_a - h_b| \le 2|a-b|$. If two of the discs in (4.1) intersect, then

$$|a-b| < 10s(h_a+h_b), \qquad \frac{h_b}{h_a} < \frac{1+20s}{1-20s}.$$

For $s < 1/80$, the ratio h_b/h_a is less than 2, so

$$\rho_\Delta(a,b) \le \frac{|a-b|}{h_a - |a-b|} < \frac{30s}{1-30s}.$$

The right-hand side is smaller than the separation constant when s is sufficiently small. Finally, $|h(\eta) - h_a| \le 2|\eta - a| < 20sh_a$, which gives (4.2) after one further reduction of s.

The separation of A implies local finiteness. Since Δ is σ-compact, the set A is at most countable. If A is finite, the sum below is finite. Otherwise fix an enumeration $A = \{a_1, a_2, \ldots\}$ and consider the partial sums $\mathcal{P}_N(\zeta) = \sum_{j=1}^N \sqrt{h_{a_j}} G(\zeta, a_j)$. In either case, write

$$\mathcal{P}(\zeta) = \sum_{a\in A} \sqrt{h_a}\, G(\zeta, a), \tag{4.3}$$

where the right-hand side is the monotone limit of $\mathcal{P}_N$ when A is infinite. If $\zeta \notin \overline{B(a, sh_a)}$, the function $G(\zeta,\cdot)$ is positive and harmonic on $B(a, sh_a)$. Its mean-value identity and (4.2) imply

$$\sqrt{h_a}\, G(\zeta,a) = \frac{1}{\pi s^2 h_a^{3/2}} \int_{B(a,sh_a)} G(\zeta,\eta)\, dA(\eta) \le C_s \int_{B(a,sh_a)} G(\zeta,\eta) h(\eta)^{-3/2}\, dA(\eta). \tag{4.4}$$

All constants below may depend on s, but are independent of a and of the parameter ε introduced in Step 2.

If ζ lies outside every $B(a, 2sh_a)$, summation of (4.4) over any finite partial sum of the disjoint averaging discs gives a uniform bound. Hence the sum in (4.3) is finite and satisfies

$$0 \le \mathcal{P}(\zeta) \le C\sqrt{h(\zeta)}. \tag{4.5}$$

For a fixed $a \in A$ and $\zeta \in B(a, 4sh_a)$, all the averaging discs centred at $b \ne a$ avoid ζ. Indeed,

$$|\zeta - b| \ge |a-b| - |\zeta - a| \ge 6sh_a + 10sh_b > sh_b.$$

Applying (4.4) to the terms indexed by $b \in A \setminus \{a\}$ and then taking the monotone limit shows that the omitted-pole series satisfies

$$\mathcal{P}_a(\zeta) := \sum_{\substack{b\in A\\ b\ne a}} \sqrt{h_b}\, G(\zeta,b) \le C\sqrt{h(\zeta)} \le C\sqrt{h_a} \qquad \big(\zeta \in B(a, 4sh_a)\big). \tag{4.6}$$

Suppose that A is infinite. On every relatively compact disc $U \Subset \Delta \setminus A$, the functions $\mathcal{P}_N$ are positive harmonic and increase with N. Estimates (4.5) and (4.6), with the single possible nearby summand treated separately, bound $\mathcal{P}_N$ at one point of U, uniformly in N. Harnack's convergence theorem therefore gives locally uniform convergence on U, and hence on $\Delta \setminus A$. Applied to the partial sums over $A \setminus \{a\}$, Harnack's convergence theorem and the bound (4.6) show that $\mathcal{P}_a$ is harmonic on $B(a, 4sh_a)$. These conclusions are immediate when A is finite. The interior gradient estimate for harmonic functions now gives

$$|\nabla \mathcal{P}_a(\zeta)| \le C h_a^{-1/2} \qquad \big(\zeta \in B(a, 2sh_a)\big). \tag{4.7}$$

Step 2: constant neighbourhoods of the points of A. Choose a constant $c_0 > 0$ such that $c_0 \log 2 > 16$. For a positive number $\varepsilon < s/2$, to be decreased below, write

$$\mu = c_0\varepsilon, \qquad v(\zeta) = \operatorname{Re}\zeta - \mu\mathcal{P}(\zeta), \qquad B_a = B(a, \varepsilon h_a), \quad B_a^+ = B(a, 2\varepsilon h_a).$$

The function v is harmonic away from A and extends subharmonically across each a with value $-\infty$: near a,

$$v(\zeta) = W_a(\zeta) + \mu\sqrt{h_a}\log\rho_\Delta(\zeta, a), \qquad W_a = \operatorname{Re}\zeta - \mu\mathcal{P}_a, \tag{4.8}$$

where W_a is harmonic. The discs B_a^+ are pairwise disjoint. Estimate (4.7) implies

$$\operatorname{osc}_{B_a^+} W_a \le 4\varepsilon h_a + C\mu\varepsilon\sqrt{h_a}. \tag{4.9}$$

Here $\operatorname{osc}_E f = \sup_E f - \inf_E f$. The identity $1 - \bar{a}\zeta = h_a - \bar{a}(\zeta - a)$ gives

$$\begin{aligned} \log\rho_\Delta(\zeta, a) &\le \log\varepsilon - \log(1-\varepsilon) && (\zeta \in \partial B_a), \\ \log\rho_\Delta(\zeta, a) &\ge \log(2\varepsilon) - \log(1+2\varepsilon) && (\zeta \in \partial B_a^+). \end{aligned}$$

For $\varepsilon < 1/4$, subtraction and (4.9) yield

$$\begin{aligned} \min_{\partial B_a^+} v - \max_{\partial B_a} v &\ge \mu\sqrt{h_a}\big(\log 2 + \log(1-\varepsilon) - \log(1+2\varepsilon)\big) - 4\varepsilon h_a - C\mu\varepsilon\sqrt{h_a} \\ &\ge \varepsilon\sqrt{h_a}\big(c_0\log 2 - 4 - C'c_0\varepsilon\big) > 0, \end{aligned} \tag{4.10}$$

after decreasing ε. In the second inequality, $h_a \le \sqrt{h_a}$ was used. The final choice of ε is uniform over A.

For each a, take

$$c_a = \frac{1}{2}\left(\max_{\partial B_a} v + \min_{\partial B_a^+} v\right).$$

The maximum principle, applied across the negative logarithmic pole of v, gives $v < c_a$ on $\overline{B_a}$. On a neighbourhood of ∂B_a^+, continuity and (4.10) give $v > c_a$. Hence the formula

$$u(\zeta) = \begin{cases} \max\{v(\zeta), c_a\}, & \zeta \in B_a^+,\ a \in A, \\ v(\zeta), & \zeta \notin \displaystyle\bigcup_{a\in A} B_a^+, \end{cases}$$

has equal expressions on an open collar of every outer circle. The discs are locally finite in Δ: if B_a^+ meets $\{|\zeta| \le r\}$, where $r < 1$, then

$$|a| - r \le 2\varepsilon h_a \le 4\varepsilon(1 - |a|), \qquad |a| \le \frac{r + 4\varepsilon}{1 + 4\varepsilon} < 1,$$

and only finitely many points of A lie in the closed disc of radius $(r+4\varepsilon)/(1+4\varepsilon) < 1$. Their interiors are disjoint. The function u is therefore continuous and subharmonic, and it equals c_a on B_a.

Step 3: boundedness, harmonicity near 0*, and a nonzero differential at* 0. Outside $\bigcup_a B_a^+$, a point ζ either lies outside every $B(a, 2sh_a)$, in which case (4.5) applies, or belongs to a unique $B(a, 2sh_a)$. In the second case,

$$\rho_\Delta(\zeta, a) \ge \frac{2\varepsilon h_a}{h_a + |\zeta - a|} \ge \frac{2\varepsilon}{1 + 2s}.$$

The own-pole term in (4.3) is then at most $C\sqrt{h(\zeta)}(1 + |\log\varepsilon|)$, and its remainder has the bound (4.6). Thus

$$0 \le \mathcal{P}(\zeta) \le C(1 + |\log\varepsilon|)\sqrt{h(\zeta)} \qquad \left(\zeta \notin \bigcup_a B_a^+\right).$$

On the two circles used to select c_a, the identities (4.8) and the bound (4.6) imply

$$|v(\zeta) - \operatorname{Re} a| \le 2\varepsilon h_a + C\mu(1 + |\log\varepsilon|)\sqrt{h_a}. \tag{4.11}$$

Their maximum and minimum, and hence their midpoint c_a, satisfy (4.11). Since $v(\zeta) \le \operatorname{Re}\zeta$, the values on B_a^+ obey

$$c_a \le u(\zeta) \le \max\{\operatorname{Re}\zeta, c_a\}.$$

The estimate $\varepsilon h_a \le (\mu/c_0)\sqrt{h_a}$ and (4.2) consequently give, both inside and outside the modified discs,

$$|u(\zeta) - \operatorname{Re}\zeta| \le C\mu(1 + |\log\varepsilon|)\sqrt{1-|\zeta|^2}. \tag{4.12}$$

In particular, u is bounded.

Local finiteness gives a neighbourhood of 0 that meets A in only finitely many points. Since $0 \notin A$, a smaller neighbourhood is disjoint from A, and hence $d_0 := \inf_{a\in A}|a| > 0$. The series $\mathcal{P}$ is harmonic on a neighbourhood of the closed disc $\{|\zeta| \le d_0/2\}$. It is independent of ε. Decrease ε once more so that $2\varepsilon < d_0/4$ and $\mu|\nabla\mathcal{P}(0)| < 1/2$. Every B_a^+ then avoids $\{|\zeta| < d_0/2\}$, and

$$u = \operatorname{Re}\zeta - \mu\mathcal{P} \quad \text{near } 0, \qquad |du(0)| \ge 1 - \mu|\nabla\mathcal{P}(0)| > \frac{1}{2}.$$

Estimates (4.9), (4.10), and (4.12) remain valid because ε has only been decreased. Finally, $\rho_\Delta(\zeta, a) < \sigma$ implies

$$|\zeta - a| \le \frac{\sigma}{1-\sigma} h_a.$$

Any $\sigma > 0$ with $\sigma/(1-\sigma) < \varepsilon$ therefore satisfies $D_{\mathrm{ph}}(a,\sigma) \subset B_a$ for every $a \in A$. □

For the polynomial P in (1.2), write

$$\begin{aligned} &Q = P + 1 = z^5 - w^2, \qquad C = \{Q = 1\}, \\ &D = \{\lambda \in \mathbb{C} : |\lambda - 1| < 1/4\}, \qquad U = Q^{-1}(D). \end{aligned} \tag{4.13}$$

The domains used below are

$$E_R = \{(z,w) : |z|^2 + |w|^2 > R^2\}, \qquad Y_R = U \cup E_R. \tag{4.14}$$

Proposition 4.2. *There is $R > 2$ such that every holomorphic map $f = (f_1, f_2) : \mathbb{C} \to Y_R$ satisfying $f(0) = (1,0)$ has $f_2 \equiv 0$.*

Proof. Step 1: the affine curve, its compactification, and the universal covering. The curve C is smooth: the equations $Q_z = 5z^4 = 0$ and $Q_w = -2w = 0$ have the unique common solution $(0,0)$, which does not belong to C. Its projection onto the z-plane is the double covering $w^2 = z^5 - 1$. Continuation around any one of the five simple roots interchanges the two sheets, so C is connected. The smooth compactification $\overline{C}$ has exactly one point q at infinity. A coordinate t at this point is

$$z = t^{-2}, \qquad w = t^{-5}\sqrt{1 - t^{10}}, \tag{4.15}$$

where the square root equals 1 at $t = 0$. For sufficiently small $|t|$, the two choices of a square root of z^{-1} differ by $t \mapsto -t$ and yield opposite values of w; thus each point of the end has a unique parameter t. This chart adjoins a single point and makes the compactification smooth. The covering $\overline{C} \to \mathbb{P}^1$, $(z,w) \mapsto z$, has simple branching at the five roots of unity and at infinity, where $1/z = t^2$. The Riemann–Hurwitz formula gives

$$2g(\overline{C}) - 2 = 2(-2) + 6 = 2, \qquad g(\overline{C}) = 2.$$

By the uniformization theorem, $\overline{C}$ has a holomorphic universal covering

$$\varpi : \Delta \longrightarrow \overline{C}, \qquad \varpi(0) = p = (1,0).$$

Its group Γ of covering transformations acts freely on Δ by pseudohyperbolic isometries. The set

$$A = \varpi^{-1}(q)$$

is one Γ-orbit and is uniformly pseudohyperbolically separated. Indeed, for one $a_0 \in A$, discreteness of the orbit gives $d_0 := \inf_{\gamma\neq 1}\rho_\Delta(a_0, \gamma a_0) > 0$. If $a = \alpha a_0$ and $b = \beta a_0$ are distinct points of A, then invariance of ρ_Δ gives $\rho_\Delta(a,b) = \rho_\Delta(a_0, \alpha^{-1}\beta a_0) \ge d_0$. Since

$0 \notin A$, Lemma 4.1 supplies a bounded continuous subharmonic function u, constants c_a, and a common radius $\sigma > 0$ such that

$$u|_{D_{\mathrm{ph}}(a,\sigma)} = c_a, \qquad u \text{ is harmonic near } 0, \quad du(0) \neq 0. \tag{4.16}$$

Choose an evenly covered disc W about q in $\overline{C}$ so small that its lifts satisfy

$$\varpi^{-1}(W) = \coprod_{a\in A} \widetilde{W}_a, \qquad a \in \widetilde{W}_a \subset D_{\mathrm{ph}}(a,\sigma).$$

Choose one lift $a_0 \in A$ and first shrink W so that $\widetilde{W}_{a_0} \subset D_{\mathrm{ph}}(a_0,\sigma)$. Since A is a single Γ-orbit and every element of Γ preserves ρ_Δ, the corresponding inclusion then holds for every $a \in A$.

Step 2: the product neighbourhood and its exterior overlap. The functions $\lambda^{1/5}$ and $\lambda^{1/2}$ have holomorphic branches on a neighbourhood of $\overline{D}$, with value 1 at $\lambda = 1$. The map

$$\Phi : C \times D \longrightarrow U, \qquad \Phi((a,b),\lambda) = (\lambda^{1/5}a, \lambda^{1/2}b) \tag{4.17}$$

is biholomorphic. In fact $Q(\Phi((a,b),\lambda)) = \lambda$, and its inverse is

$$\Phi^{-1}(z,w) = (\pi(z,w), Q(z,w)), \qquad \pi(z,w) = \big(Q(z,w)^{-1/5}z, Q(z,w)^{-1/2}w\big).$$

The overlap $U \cap E_R$ can be described explicitly near infinity. In the coordinates of (4.15) and (4.17), its squared Euclidean norm is

$$N(r,\theta,\lambda) = |\lambda|^{2/5}r^{-4} + |\lambda|r^{-10}|1 - r^{10}e^{10i\theta}|, \qquad t = re^{i\theta}. \tag{4.18}$$

For $0 < r < 1$,

$$\frac{d}{dr}\big(r^{-10}|1 - r^{10}e^{10i\theta}|\big) = -\frac{10r^{-11}(1 - r^{10}\cos 10\theta)}{|1 - r^{10}e^{10i\theta}|} < 0. \tag{4.19}$$

The first term of (4.18) also has negative radial derivative. Moreover, $N(r,\theta,\lambda) \to +\infty$ as $r \downarrow 0$ uniformly for $\theta \in \mathbb{R}/2\pi\mathbb{Z}$ and $\lambda \in \overline{D}$.

Take a fixed end chart $0 < |t| < r_*$ with $r_* < 1$. Its complement in C is compact. The image of that complement times $\overline{D}$ under (4.17) is bounded. Consequently, for every sufficiently large R, the whole overlap $U \cap E_R$ lies in the end chart. Equations (4.18)–(4.19) then give a unique positive function $r_R(\theta,\lambda) < r_*$ such that

$$\begin{aligned} N(r_R(\theta,\lambda),\theta,\lambda) &= R^2, \\ U \cap E_R &= \{(t,\lambda) : 0 < |t| < r_R(\arg t,\lambda),\ \lambda \in D\}. \end{aligned} \tag{4.20}$$

The implicit function theorem makes r_R smooth. For every $0 < b < r_*$, the function $N(b,\theta,\lambda)$ is bounded on $(\mathbb{R}/2\pi\mathbb{Z}) \times \overline{D}$; hence

$$\sup_{\theta,\lambda} r_R(\theta,\lambda) \longrightarrow 0 \qquad (R \to \infty).$$

Choose $R > 2$ so large that the description (4.20) holds and that

$$\pi(U \cap E_R) \subset W \setminus \{q\}.$$

The map

$$(s, e^{i\theta}, \lambda) \longmapsto \big(sr_R(\theta,\lambda)e^{i\theta}, \lambda\big)$$

is a homeomorphism from $(0,1) \times S^1 \times D$ onto the overlap. It follows that $U \cap E_R$ is connected and that its fundamental group is generated by the class of the loop $\theta \mapsto (t_0 e^{i\theta}, 1)$, $0 \le \theta \le 2\pi$, for any $0 < t_0 < \min_\theta r_R(\theta,1)$. Under the projection to C, this class is represented by a primitive peripheral loop about q. After joining its base point to the fixed base point of C, denote the resulting class in $\pi_1(C)$ by ℓ_q.

Step 3: the fundamental group and the lifted open cover. The exterior E_R deformation retracts onto a three-sphere and is simply connected. Van Kampen's theorem [18, Theorem 1.20] and $U \cong C \times D$ therefore give

$$\pi_1(Y_R) \cong \pi_1(C)/\langle\langle \ell_q \rangle\rangle \cong \pi_1(\overline{C}) = \Gamma,$$

where $\langle\langle \ell_q \rangle\rangle$ denotes the normal closure of ℓ_q in $\pi_1(C)$. The homomorphism induced by $U \hookrightarrow Y_R$ is the quotient homomorphism induced by $C \hookrightarrow \overline{C}$.

Let

$$\Pi : \widetilde{Y}_R \longrightarrow Y_R$$

be the universal covering, with its lifted complex structure. The monodromy description of covering spaces [22, Chapter 3, Sections 4 and 7], applied to the surjection $\pi_1(U) \to \pi_1(Y_R)$, shows that $\widetilde{U} = \Pi^{-1}(U)$ is connected and that the image of $\pi_1(\widetilde{U})$ in $\pi_1(U)$ is $\ker[\pi_1(C) \to \pi_1(\overline{C})]$. The image of $\pi_1(\Delta \setminus A)$ under the covering $\varpi : \Delta \setminus A \to C$ is the same subgroup: a loop in C lifts to a closed loop in $\Delta \setminus A$ if and only if it is null-homotopic in $\overline{C}$. The classification of connected coverings [18, Theorem 1.38] consequently gives a biholomorphic identification

$$\widetilde{U} \cong (\Delta \setminus A) \times D, \qquad \Pi(\zeta, \lambda) = \Phi(\varpi(\zeta), \lambda). \tag{4.21}$$

Both covering maps are local biholomorphisms, so the covering isomorphism in (4.21) is biholomorphic.

Every connected component V of $\Pi^{-1}(E_R)$ maps biholomorphically onto E_R. In particular,

$$V \cap \widetilde{U} = \big(\Pi|_V\big)^{-1}(E_R \cap U) \tag{4.22}$$

is connected and is a single copy of $E_R \cap U$.

Step 4: a bounded plurisubharmonic function on the cover. Each connected set $V \cap \widetilde{U}$ in (4.22) has its first coordinate in one $\widetilde{W}_a \setminus \{a\}$. The formula

$$\widetilde{u} = \begin{cases} u(\zeta), & \text{on } \widetilde{U} \cong (\Delta \setminus A) \times D, \\ c_a, & \text{on a component } V \text{ whose overlap lies over } \widetilde{W}_a, \end{cases} \tag{4.23}$$

therefore has identical values on every open overlap, by (4.16). The components V are pairwise disjoint. On $\widetilde{U}$, the first expression is plurisubharmonic because the first-coordinate projection is holomorphic and u is subharmonic. On every V, the second expression is constant. Hence $\widetilde{u}$ is a bounded continuous plurisubharmonic function on $\widetilde{Y}_R$. Its boundedness follows from the boundedness of u, since each c_a is a value of u.

Step 5: the restriction on entire curves. For a holomorphic map $f : \mathbb{C} \to Y_R$ with $f(0) = p$, the covering-space lifting criterion [18, Proposition 1.33] and simple connectedness of $\mathbb{C}$ give a holomorphic lift

$$\widetilde{f} : \mathbb{C} \longrightarrow \widetilde{Y}_R, \qquad \widetilde{f}(0) = (0, 1)$$

in the coordinates of (4.21). The function $\widetilde{u} \circ \widetilde{f}$ is bounded and subharmonic on $\mathbb{C}$, so it is constant. Near 0 in the source, write

$$\widetilde{f}(s) = (\xi(s), \lambda(s)), \qquad \xi(0) = 0, \quad \lambda(0) = 1.$$

It follows that $u(\xi(s))$ is constant. On a sufficiently small simply connected neighbourhood of 0 in Δ, (4.16) gives

$$u = \operatorname{Re} H, \qquad H \text{ holomorphic}, \qquad H' \neq 0.$$

After restricting the source neighbourhood accordingly,

$$0 = \frac{\partial}{\partial s} u(\xi(s)) = \frac{1}{2} H'(\xi(s)) \xi'(s).$$

Thus $\xi \equiv 0$ near $s = 0$. Equations (4.17) and (4.21) give

$$f(s) = \Phi(p, \lambda(s)) = (\lambda(s)^{1/5}, 0) \quad \text{near } s = 0.$$

The second component of f vanishes on a nonempty open set; the identity theorem gives $f_2 \equiv 0$ on $\mathbb{C}$. □

We now use Proposition 4.2 to construct explicit strongly pseudoconvex compact sets whose complements are not Oka. For $P(z, w) = z^5 - w^2 - 1$, the function occurring in (1.2) is

$$\psi(z, w) = |z|^2 + |w|^2 + |P(z, w)|^{-2} \quad \text{on } M = \mathbb{C}^2 \setminus \{P = 0\}.$$

Its large sublevel sets are regular without any choice of a generic parameter.

Lemma 4.3. *Every critical value of ψ is strictly less than 6.*

Proof. At a critical point of ψ, the complex derivative equations are

$$\bar{z} = 5cz^4, \qquad \bar{w} = -2cw, \qquad c = \frac{1}{P^2 \overline{P}}. \tag{4.24}$$

If $w \neq 0$, then $|c| = 1/2$ and $|P|^3 = 2$. The first equation gives either $z = 0$ or $|z|^3 = 2/5$, so $|z| < 1$. Since $w^2 = z^5 - 1 - P$,

$$|w|^2 \leq |z|^5 + 1 + |P| < 1 + 1 + 2 = 4, \qquad \psi < 1 + 4 + 1 = 6.$$

If $w = 0$ and $z \neq 0$, write $r = |z|$. The first equation in (4.24) becomes

$$|z^5 - 1|^3 = 5r^3. \tag{4.25}$$

For $r \leq 1/2$,

$$|z^5 - 1|^3 \geq (1 - r^5)^3 \geq (31/32)^3 > 5/8 \geq 5r^3,$$

which contradicts (4.25). For $r \geq 2$,

$$|z^5 - 1|^3 \geq (r^5 - 1)^3 \geq r^{15}/8 > 5r^3,$$

again a contradiction. Hence $1/2 < r < 2$, and

$$|P|^{-2} = 5^{-2/3} r^{-2} < 4 \cdot 5^{-2/3} < 2, \qquad \psi = r^2 + |P|^{-2} < 6.$$

At the only remaining point, $(z, w) = (0, 0)$, the value is $\psi = 1$. □

Proof of Theorem 1.3. Choose $R > 2$ as in Proposition 4.2 and take

$$T_0 = R^2 + 16.$$

Fix $T > T_0$.

(i) The function ψ is a real-analytic proper exhaustion of M. In fact, its sublevel set satisfies

$$K_T \subset \{(z, w) : |z|^2 + |w|^2 \leq T,\ |P(z, w)| \geq T^{-1/2}\}. \tag{4.26}$$

The right-hand side is compact in M, and K_T is closed there. At $Z = (z, w) \in M$ and $v = (v_1, v_2) \in \mathbb{C}^2$,

$$\begin{aligned} \mathcal{L}_\psi(Z; v) &= |v_1|^2 + |v_2|^2 + |d(1/P)_Z(v)|^2 \\ &= |v_1|^2 + |v_2|^2 + \frac{|5z^4 v_1 - 2wv_2|^2}{|P(Z)|^4}. \end{aligned}$$

Thus ψ is strictly plurisubharmonic. By Lemma 4.3, $T > 6$ is a regular value. Consequently $\{\psi = T\}$ is a real-analytic strongly pseudoconvex hypersurface and

$$K_T = \overline{\Omega_T}, \qquad \Omega_T = \{Z \in M : \psi(Z) < T\}.$$

The last equality follows in local regular-level coordinates: every point of $\{\psi = T\}$ is approached from $\{\psi < T\}$.

The manifold M is path connected. For $x, y \in M$, the restriction of P to the complex affine line through x, y is a nonzero polynomial, since it is nonzero at x. Its zero set on that line is finite, and the complement of a finite set in $\mathbb{C}$ is path connected.

Given $x, y \in \Omega_T$, choose $\max\{6, \psi(x), \psi(y)\} < S < T$ and a path $\gamma : [0,1] \to M$ from x to y. Let

$$B = \max\left\{S, \max_{t\in[0,1]} \psi(\gamma(t))\right\}.$$

If $B > S$, then $\psi^{-1}([S, B])$ is compact and contains no critical point. The no-critical-values deformation theorem [23, Theorem 3.1] therefore deforms $\{\psi \le B\}$ onto $\{\psi \le S\}$ while fixing $\{\psi \le S\}$ pointwise. Applying the deformation to γ gives a path from x to y in $\{\psi \le S\} \subset \Omega_T$. If $B = S$, the original path γ already lies in $\{\psi \le S\} \subset \Omega_T$. Hence Ω_T is connected.

For $|z| = 2$ and $w = 0$,

$$\psi(z, 0) = 4 + \frac{1}{|z^5 - 1|^2} \le 4 + \frac{1}{31^2} < 5 < T. \tag{4.27}$$

The circle $\{(z, 0) : |z| = 2\}$ therefore belongs to K_T. Since $P(1, 0) = 0$, the point $p = (1, 0)$ does not belong to K_T.

(ii) Lemma 2.10, applied to $K_T \subset \mathbb{C}^2$, proves that X_T is connected. For $Z \notin Y_R$, equations (4.13) and (4.14) give

$$|Z|^2 \le R^2, \qquad |P(Z)| \ge 1/4, \qquad \psi(Z) \le R^2 + 16 = T_0 < T.$$

It follows that

$$X_T \subset Y_R. \tag{4.28}$$

If $f = (f_1, f_2) : \mathbb{C} \to X_T$ is holomorphic and $f(0) = p$, Proposition 4.2 and (4.28) give $f_2 \equiv 0$. By (4.27), the connected set $f_1(\mathbb{C})$ avoids $\{|z| = 2\}$ and contains 1. Thus $|f_1| < 2$ on $\mathbb{C}$. Liouville's theorem yields $f_1 \equiv 1$, so $f \equiv p$.

Lemma 2.1 now shows that X_T is not Oka. □

5. Counterexamples on complex surfaces and a necessary Levi condition

The function $\widetilde{u}$ in (4.23) is constant on each connected component of $\Pi^{-1}(E_R)$. To transfer this construction to a complex surface M, we identify the bounded set $Y_{R,S} = Y_R \cap B_S$ with a coordinate neighbourhood $j(Y_{R,S}) \subset M$ and join it to $E_M = M \setminus j(\overline{B}_R)$. The covering of the resulting open set Y_M is chosen so that its restriction over $j(Y_{R,S})$ is the universal covering of $Y_{R,S}$ and its restriction to every connected component over E_M is biholomorphic to E_M. These two properties allow $\widetilde{u}$ to be extended by constants over the preimage of E_M.

Lemma 5.1. *Let $d \ge 2$ and $0 < a < b$. The following assertions hold.*

(i) *If N is a connected real manifold of dimension d and $j : B_b \to N$ is a coordinate embedding, then $N \setminus j(\overline{B}_a)$ is path connected.*

(ii) *If $Y \subset \mathbb{R}^d$ is a connected open set containing $\mathbb{R}^d \setminus \overline{B}_a$, then $Y \cap B_b$ is path connected.*

Proof. (i) The shell

$$C = j(B_b \setminus \overline{B}_a) \cong (a, b) \times S^{d-1}$$

is connected and belongs to one connected component E_0 of $E = N \setminus j(\overline{B}_a)$. Suppose another component E_1 exists. Components of the manifold E are both open and closed in E, so

$$\overline{E}_1^N \setminus E_1 \subset j(\partial B_a).$$

Every point of $j(\partial B_a)$ has a neighbourhood W in $j(B_b)$ with $W \cap E \subset C$. Since $E_1 \cap C$ is empty, no such point belongs to $\overline{E}_1^N$. Thus E_1 is both open and closed in N, contrary

to the connectedness of N. Hence E is connected, and local path connectedness makes it path connected.

(ii) For $x, y \in Y \cap B_b$, choose

$$\max\{a, |x|, |y|\} < t < b.$$

The radial map

$$r_t(v) = \begin{cases} v, & |v| \le t, \\ tv/|v|, & |v| > t \end{cases}$$

is continuous, fixes x and y, and satisfies

$$r_t(Y) \subset Y \cap \overline{B}_t \subset Y \cap B_b.$$

Indeed, an altered point has norm $t > a$ and therefore still belongs to Y. A path in Y from x to y, composed with r_t, is the required path in $Y \cap B_b$. □

Theorem 5.2. *Let M be a connected complex surface, $q \in M$, and $V \subset M$ be an open neighbourhood of q. There is a compact set $K \Subset V$ with the following properties.*

(i) *K is the closure of a connected domain, $q \notin K$, and ∂K is real analytic and strongly pseudoconvex.*
(ii) *$M \setminus K$ is connected, and every holomorphic map $f : \mathbb{C} \to M \setminus K$ with $f(0) = q$ is constant. Consequently $M \setminus K$ is not Oka.*

Proof. Retain $R > 2$, $Y_R = U \cup E_R$, the universal covering $\Pi : \widetilde{Y}_R \to Y_R$, and the bounded continuous plurisubharmonic function $\widetilde{u}$ from the proof of Proposition 4.2. Thus

$$\pi_1(Y_R) \cong \Gamma = \pi_1(\overline{C}), \qquad \widetilde{u}|_{\widetilde{E}} \text{ is constant for every component } \widetilde{E} \text{ of } \Pi^{-1}(E_R).$$

The distinguished point is $p = (1, 0)$. Choose the lift with coordinates $(0, 1)$ in (4.21). On a neighbourhood of this lift, we have

$$\widetilde{u}(\zeta, \lambda) = u(\zeta), \qquad u = \operatorname{Re} H, \qquad H'(0) \neq 0, \tag{5.1}$$

where H is holomorphic near 0.

Choose $T > R^2 + 16$ and the compact domain K_T of Theorem 1.3. Fix

$$S > \max\{R, \sqrt{T}, 2\}, \qquad Y_{R,S} = Y_R \cap B_S, \qquad \mathcal{S}_{R,S} = \{x \in \mathbb{C}^2 : R < |x| < S\}.$$

Step 1: the truncated region and its covering. Since Y_R is connected and contains E_R, Lemma 5.1(ii) shows that $Y_{R,S}$ is connected. The exact open-cover relations are

$$Y_R = Y_{R,S} \cup E_R, \qquad Y_{R,S} \cap E_R = \mathcal{S}_{R,S}.$$

The shell $\mathcal{S}_{R,S} \cong (R, S) \times S^3$ and the exterior $E_R \cong (R, \infty) \times S^3$ are simply connected. Van Kampen's theorem [18, Theorem 1.20] consequently gives an isomorphism induced by inclusion,

$$\pi_1(Y_{R,S}) \xrightarrow{\cong} \pi_1(Y_R) = \Gamma. \tag{5.2}$$

A base point in $\mathcal{S}_{R,S}$ is used in (5.2), and a fixed path in $Y_{R,S}$ joins it to p.

By the monodromy description of covering spaces [22, Chapter 3, Sections 4 and 7], the isomorphism in (5.2) implies that the preimage

$$\widetilde{Y}_{R,S} = \Pi^{-1}(Y_{R,S})$$

is connected and $\Pi|_{\widetilde{Y}_{R,S}}$ is the universal covering of $Y_{R,S}$. Each component of $\Pi^{-1}(\mathcal{S}_{R,S})$ lies in a unique component $\widetilde{E}$ of $\Pi^{-1}(E_R)$, and

$$\widetilde{E} \cap \widetilde{Y}_{R,S} = \left(\Pi|_{\widetilde{E}}\right)^{-1}(\mathcal{S}_{R,S}) \cong \mathcal{S}_{R,S}. \tag{5.3}$$

Here $\widetilde{E}$ is the component of $\Pi^{-1}(E_R)$ under consideration, and $\Pi|_{\widetilde{E}} : \widetilde{E} \to E_R$ is a biholomorphism, since E_R is simply connected. In particular, the function $\widetilde{u}|_{\widetilde{Y}_{R,S}}$ is constant on each lifted shell, with the constant belonging to its unique component $\widetilde{E}$.

Step 2: the ambient coordinate region and the new covering. A holomorphic coordinate map $\kappa : V_0 \to \kappa(V_0)$ can be chosen with

$$q \in V_0 \subset V, \qquad \kappa(q) = 0.$$

For sufficiently small $\delta > 0$, the formula

$$j(x) = \kappa^{-1}\big(\delta(x-p)\big), \qquad x \in B_{S+1},$$

has image in V_0 and satisfies $j(\overline{B}_S) \Subset V_0$ and $j(p) = q$. Indeed, a Euclidean ball about 0 lies in $\kappa(V_0)$, and the bounded set $\{x - p : |x| \le S+1\}$ fits into that ball after multiplication by a sufficiently small δ.

The open sets

$$E_M = M \setminus j(\overline{B}_R), \qquad Y_M = j(Y_{R,S}) \cup E_M \tag{5.4}$$

satisfy

$$j(Y_{R,S}) \cap E_M = j(\mathcal{S}_{R,S}).$$

Lemma 5.1(i) makes E_M connected. Thus both sets in (5.4), and their nonempty intersection, are path connected. Since the intersection is simply connected, van Kampen's theorem gives

$$\pi_1(Y_M) \cong \Gamma * \Lambda, \qquad \Lambda = \pi_1(E_M).$$

Here $*$ denotes the free product. Under this identification, the inclusions of $j(Y_{R,S})$ and E_M induce the two free-factor inclusions.

The homomorphism

$$\eta : \Gamma * \Lambda \longrightarrow \Gamma, \qquad \eta|_\Gamma = \mathrm{id}_\Gamma, \quad \eta|_\Lambda = 1,$$

is surjective. The subgroup $\ker \eta$ determines a connected covering

$$\widehat{\Pi} : \widehat{Y}_M \longrightarrow Y_M \tag{5.5}$$

by the covering classification [18, Theorem 1.38]. The lifted complex structure makes $\widehat{\Pi}$ a local biholomorphism.

By the monodromy description [22, Chapter 3, Sections 4 and 7], the fibre of (5.5) is the $(\Gamma * \Lambda)$-set $(\Gamma * \Lambda)/\ker\eta \cong \Gamma$, with the action induced by η. Its restriction to $\pi_1(j(Y_{R,S})) = \Gamma$ is the left-translation action of Γ on itself. Hence

$$\widehat{Y}_{R,S} = \widehat{\Pi}^{-1}(j(Y_{R,S}))$$

is connected and $\widehat{\Pi}|_{\widehat{Y}_{R,S}}$ is the universal covering of $j(Y_{R,S})$. There is consequently a biholomorphism

$$I : \widehat{Y}_{R,S} \longrightarrow \widetilde{Y}_{R,S}, \qquad \Pi \circ I = j^{-1} \circ \widehat{\Pi}, \tag{5.6}$$

which can be chosen to carry the selected lift of q to the selected lift of p.

In contrast, $\eta|_\Lambda = 1$ makes every Λ-orbit in the fibre a singleton. The same monodromy result shows that, for every component $\widehat{E}$ of $\widehat{\Pi}^{-1}(E_M)$,

$$\widehat{\Pi}|_{\widehat{E}} : \widehat{E} \longrightarrow E_M \quad \text{is a biholomorphism.} \tag{5.7}$$

In particular,

$$\widehat{E} \cap \widehat{Y}_{R,S} = \big(\widehat{\Pi}|_{\widehat{E}}\big)^{-1}\big(j(\mathcal{S}_{R,S})\big) \tag{5.8}$$

is connected and is one copy of the shell.

Step 3: construction of a bounded plurisubharmonic function. For each component $\widehat{E}$ in (5.7), its overlap in (5.8) is carried by I to one component of $\Pi^{-1}(\mathcal{S}_{R,S})$. Equation (5.3)

places that component in a unique component $\widetilde{E}$ of $\Pi^{-1}(E_R)$. Let $c_{\widehat{E}}$ be the constant value of $\widetilde{u}$ on $\widetilde{E}$. The formulas

$$\widehat{u} = \begin{cases} \widetilde{u} \circ I, & \text{on } \widehat{Y}_{R,S}, \\ c_{\widehat{E}}, & \text{on each component } \widehat{E} \text{ of } \widehat{\Pi}^{-1}(E_M) \end{cases}$$

agree on every open overlap. The components $\widehat{E}$ are disjoint. Both types of local expression are continuous and plurisubharmonic, and all constants $c_{\widehat{E}}$ are values of the bounded function $\widetilde{u}$. Therefore $\widehat{u}$ is a bounded continuous plurisubharmonic function on $\widehat{Y}_M$. Near the chosen lift of q, it is given by (5.1).

Step 4: the compact set and connectedness. Set

$$K = j(K_T).$$

The inclusion $K_T \subset B_S$ follows from (4.26) and $S > \sqrt{T}$. Thus $K \Subset V$. Theorem 1.3(i) and the biholomorphism j show that K is the closure of a connected domain with real-analytic strongly pseudoconvex boundary. Moreover, $p \notin K_T$ gives $q \notin K$. This proves (i).

For connectedness in (ii), choose

$$\sqrt{T} < S_0 < S.$$

The Euclidean complement $X_T = \mathbb{C}^2 \setminus K_T$ is connected by Theorem 1.3(ii) and contains $\mathbb{C}^2 \setminus \overline{B}_{\sqrt{T}}$. Lemma 5.1(ii) therefore makes $X_T \cap B_S$ connected. The exact open cover

$$M \setminus K = j(X_T \cap B_S) \cup \big(M \setminus j(\overline{B}_{S_0})\big)$$

has connected members by Lemma 5.1(i). Their intersection is the nonempty connected shell $j(\{S_0 < |x| < S\})$. Hence $M \setminus K$ is connected.

The containment (4.28) also gives

$$M \setminus K \subset Y_M. \tag{5.9}$$

Indeed, a point outside $j(\overline{B}_R)$ belongs to E_M. A point in $j(\overline{B}_R) \setminus K$ has coordinate $x \in X_T \cap \overline{B}_R \subset Y_R \cap B_S = Y_{R,S}$.

Step 5: entire curves and confinement to the coordinate region. Let $f : \mathbb{C} \to M \setminus K$ be holomorphic with $f(0) = q$. By (5.9), it is a map to Y_M. The lifting criterion [18, Proposition 1.33] and $\pi_1(\mathbb{C}) = 0$ give a lift

$$\widehat{f} : \mathbb{C} \longrightarrow \widehat{Y}_M$$

through the selected lift of q. The lift is holomorphic because $\widehat{\Pi}$ is a local biholomorphism. The bounded subharmonic function $\widehat{u} \circ \widehat{f}$ is constant on $\mathbb{C}$.

On a sufficiently small disc $D_0 \subset \mathbb{C}$ centred at 0, the map $I \circ \widehat{f}$ takes values in the coordinates (ζ, λ) from (4.21); write

$$(I \circ \widehat{f})(s) = (\xi(s), \lambda(s)), \qquad \xi(0) = 0, \quad \lambda(0) = 1.$$

Equations (5.1) and (5.6) show that $u(\xi(s)) = \operatorname{Re} H(\xi(s))$ is constant on D_0. After decreasing D_0, $H'(\xi(s)) \neq 0$ for every $s \in D_0$, because $H'(0) \neq 0$. Hence

$$0 = \frac{\partial}{\partial s} u(\xi(s)) = \frac{1}{2} H'(\xi(s)) \xi'(s),$$

so $\xi \equiv 0$ on D_0. The projection formula (4.17) then gives

$$j^{-1} \circ f(s) = (\lambda(s)^{1/5}, 0) \quad (s \in D_0). \tag{5.10}$$

Let Ω be the connected component of $f^{-1}(j(B_S))$ containing 0. On this connected open set,

$$g = j^{-1} \circ f = (g_1, g_2) : \Omega \longrightarrow B_S$$

is holomorphic. Equation (5.10) and the identity theorem give $g_2 \equiv 0$ on Ω. The circle in (4.27) belongs to K_T, so

$$|g_1(s)| \neq 2 \quad (s \in \Omega), \qquad g_1(0) = 1.$$

Connectedness of Ω implies

$$|g_1(s)| < 2 \quad (s \in \Omega). \tag{5.11}$$

Consequently $f(\Omega) \subset j(\{(z,0) : |z| < 2\})$.

The set Ω is closed in $\mathbb{C}$. To verify this, suppose $s_\nu \in \Omega$ and $s_\nu \to s_* \in \mathbb{C}$. Continuity of f and compactness of $j(\{(z,0) : |z| \leq 2\})$ give

$$f(s_*) \in j(\{(z,0) : |z| \leq 2\}) \subset j(B_S).$$

There is a connected source disc D_* about s_* contained in $f^{-1}(j(B_S))$. It meets Ω for large ν, so it belongs to that same connected component. In particular, $s_* \in \Omega$. Since Ω is nonempty and open, $\Omega = \mathbb{C}$. Now g_1 is a bounded entire function by (5.11); Liouville's theorem gives $g_1 \equiv 1$. Hence $f \equiv q$. Lemma 2.1 proves the final assertions of (ii). □

We finish by proving a necessary condition on the tangential Levi form of a compact set with Oka complement. A strongly pseudoconvex boundary point of a domain obstructs the Oka property. The local peak-function construction and its extension by a negative constant are recalled by Forstnerič–Lárusson [12, p. 7]; the argument is local in the ambient manifold.

Proposition 5.3. *Let M be a complex manifold of dimension $n \geq 2$ and $K \subset M$ be a closed set with C^2 boundary. Suppose that, at a point $a \in \partial K$, a local defining function ρ on a neighbourhood U of a, with $K \cap U = \{\rho \leq 0\}$, has negative-definite tangential Levi form. Then there is a nonempty open set $W \subset M \setminus K$ such that every holomorphic map $f : \mathbb{C} \to M \setminus K$ with $f(0) \in W$ is constant. Consequently $M \setminus K$ is not Oka.*

Proof. Write $X = M \setminus K$. The hypotheses make a a strongly pseudoconvex boundary point of X, or equivalently a strongly pseudoconcave boundary point of K. The standard peak-function construction [12, p. 7], which is local and hence applies in the ambient manifold M, gives a nonempty open set $W \subset X$ and a bounded-above plurisubharmonic function φ on X that is smooth and strictly plurisubharmonic on W.

Let $f : \mathbb{C} \to X$ be holomorphic with $f(0) \in W$. The function $\varphi \circ f$ is bounded above and subharmonic on $\mathbb{C}$, so it is constant. On a sufficiently small disc D about 0, the image $f(D)$ lies in W. The chain rule gives

$$0 = \frac{\partial^2}{\partial\zeta\,\partial\overline{\zeta}}(\varphi \circ f)(\zeta) = \mathcal{L}_\varphi(f(\zeta); f'(\zeta)) \quad (\zeta \in D).$$

Strict plurisubharmonicity yields $f'|_D = 0$. Hence f is constant on D, and the identity theorem makes it constant on $\mathbb{C}$. Lemma 2.1 excludes the Oka property. □

Corollary 5.4. *Let M be a complex manifold of dimension $n \geq 2$ and $K \subset M$ be a closed set with C^2 boundary. If $M \setminus K$ is Oka, then the tangential Levi form of ∂K, computed using a local defining function ρ with $K = \{\rho \leq 0\}$ near the boundary point, has at least one nonnegative eigenvalue at every boundary point.*

Proof. A Hermitian form on the $(n-1)$-dimensional complex tangent space has no nonnegative eigenvalue exactly when it is negative definite. Proposition 5.3 excludes this alternative. □

Corollary 5.5. *Let M be a complex surface and $K \subset M$ be a closed set with C^2 boundary. If $M \setminus K$ is Oka, then ∂K is pseudoconvex with the orientation of K.*

Proof. The complex tangent space of ∂K has dimension one. Its single Levi eigenvalue is nonnegative by Corollary 5.4, which is precisely pseudoconvexity. □

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Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China

Email address: duyunheng22@mails.ucas.ac.cn